TAYLOR WESTFALL-MCCOY

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EDUCATION

University of Wisconsin–Madison — M.S. Student, Animal Sciences 2025–Present
Graduate GPA: 3.853/4.000

University of Minnesota, Twin Cities — B.S., Animal Science May 2019
Companion Animal Emphasis | Degree GPA: 3.043/4.000

RESEARCH INTERESTS

Neuroethology; behavioral neuroendocrinology; social and environmental modulation of animal behavior; animal welfare science and research ethics; quantitative, computational, and mixed-methods approaches to behavioral research.

RESEARCH EXPERIENCE

Graduate Researcher — Veterinarian Rumination Study, UW–Madison 2025–Present

- Developing a mixed-methods survey of rumination in veterinarians using the RRS-10, a five-item work-related rumination measure, and original qualitative questions; prepared methodology and supporting documentation for human-subjects review.
- Distilled more than 300 published first-person responses into meaning units and codes and collaborated on codebook refinement to develop candidate survey categories.
- Developed an R-based analysis plan grounded in published RRS-10 methods to examine brooding, reflection, and relationships with qualitative themes.

Ronald E. McNair Undergraduate Research Scholar, University of Minnesota 2017–2018

- 2018: Investigated the effect of β-hydroxy-β-methylbutyrate on proteolysis via ubiquitination in equine skeletal muscle cells using Western blot techniques.
- 2017: Investigated cold-plasma decontamination of chicken breast, romaine lettuce, and stainless-steel surfaces exposed to multidrug-resistant Salmonella enterica serovar Heidelberg.
- Applied experimental design, microbiology and molecular-biology methods, statistical analysis, scientific writing, and research presentation.

Independent Research Systems Development 2026–Present

- Developed Python and shell utilities for source-traceable research records, artifact-integrity checks, structured handoffs, and failure recovery.
- Translated research requirements into documented workflows with explicit validation criteria, uncertainty reporting, and reproducible acceptance checks.

APPLIED ANIMAL BEHAVIOR EXPERIENCE

Professional Animal Handler — Canine and Equine Performance 2019–Present

- Manage and condition teams of up to 15 dogs for high-level competition; adapt classical, operant, and second-order conditioning to individual animals and performance contexts.
- Bred, owned, and handled a Doberman Grand Champion to two Best in Specialty Show wins; placed 24th in Rally Excellent at the 2019 AKC Rally National Championship.
- First and only Black handler to guide a Doberman through all three DPCA Register of Merit components: conformation, performance, and temperament evaluation.

Intern — Hilltop Farm, Inc., Maryland May–August 2016

- Worked in warmblood breeding and training departments; assisted with foal handling, reproductive management, and performance-horse conditioning.
- Gained experience with selective breeding strategies and performance evaluation in elite sport horses.

SELECTED COURSEWORK AND QUANTITATIVE TRAINING

Graduate coursework: ZOOLOGY 620 Neuroethology Seminar (A); ZOOLOGY 335 Human/Animal Relationships: Biological and Philosophical Issues (A); STAT 303 R for Statistics I (A); STAT 304 R for Statistics II (A); AN SCI 875 Rigor in Animal Behavior Research (AB); MED HIST 558 Ethical Issues in Health Care (A).

Undergraduate coursework: ANSC 3480 Introduction to Animal Behavior (A−); ANSC 2015 Animal Welfare and Ethics (A).

HONORS AND AWARDS

- Ronald E. McNair Postbaccalaureate Achievement Program Scholar, 2017 and 2018.
- University of Minnesota Dean's List, Spring 2018.
- Undergraduate merit-based full-scholarship recipient.

RESEARCH AND TECHNICAL SKILLS

Quantitative and mixed methods: R; statistical analysis; survey design; qualitative coding; codebook development; literature synthesis; experimental design.

Laboratory: Western blotting; microbiology and molecular-biology methods; cold-plasma pathogen-decontamination research; equine skeletal-muscle-cell research.

Behavioral: Behavioral observation; classical, operant, and second-order conditioning; canine and equine performance preparation and care.

Computing: Python and shell tooling; Markdown/JSON research records; artifact verification; source-traceable workflow design.